Counselling Code : VBIT
VIGNANA BHARATHI Institute of Technology
(A UGC Autonomous Institution, Approved by AICTE, Accredited by NBA & NAAC-A Grade, Affiliated to JNTUH)

I B Tech I Semester: 2026-27

PROGRAMMING FOR PROBLEM SOLVING

PARTICIPATORY REPORT ASSESSMENT

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Branch & section	**:**	**CSE-F**
Title/ Topic of PRA	**:**	**WordSmith: Word Anagram and Text Analysis**
Name of the Faculty	**:**	**Arun G**

ABSTRACT

WordSmith is a console-based word anagram game, written in C, that tests a player's vocabulary and logical thinking. The program picks a word from a stored list, shuffles its letters to form a scrambled word, and asks the player to rearrange them into the original word. Since an anagram is a rearrangement of the same letters, two words are anagrams exactly when they contain the same letters with the same frequencies. The program uses this idea to check every guess.

This report presents the design and implementation of the game using core programming concepts: arrays and strings for storing words, functions for modular design, loops and conditional statements for game flow, and the rand() function for shuffling letters. Text analysis is built into the game, including counting the letters, vowels and consonants of a word, finding its length, and checking for palindromes and anagram pairs by comparing letter frequencies. The game also keeps score, gives hints and limits the number of attempts, with sample inputs and outputs to illustrate each feature.

Beyond entertainment, WordSmith shows how a real problem can be broken into smaller steps, solved with algorithms and coded in a structured way. The same string-handling and text-analysis techniques are used in spell checkers, search engines, word-puzzle apps and natural language processing.

Keywords: WordSmith, Word anagram, Text analysis, String handling, C programming.

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